

Counting Figures (Triangles & Squares)

COMPLETE LEARNING GUIDE

What You'll Learn

Counting-figures questions ask exactly how many triangles, squares, or rectangles exist within a given composite figure, and are deceptively hard because the smallest visible pieces are never the full answer — larger figures formed by combining adjacent smaller pieces must also be counted. This sub-module builds a fully systematic method (smallest units first, then every valid combination), applies it to the classic cevian-triangle problem, the six-pointed star, $n \times n$ square-counting grids, tilted (diamond) squares formed by midpoint lines, and composite figures that mix triangles and rectangles together — each worked to an exact, verified final count.

1. The Systematic Method: Count Smallest Units, Then All Combinations

For any counting-figures question, first identify and count the smallest, indivisible regions in the figure (the pieces that can't be split further by any drawn line), then systematically count every larger figure formed by combining 2, 3, or more of those adjacent smallest regions — the total count is the sum of the smallest-unit count plus every valid combination count, never just the smallest units alone.

- A rectangle is divided by one vertical line into 2 equal smaller rectangles. Smallest units: 2. Combinations: the 2 smaller rectangles combined recreate the original whole rectangle: 1 more. Total rectangles in the figure = $2 + 1 = 3$, not just 2.

2. Counting Triangles Formed by Multiple Lines from One Vertex

When several lines are drawn from a single vertex of a triangle to different points along the opposite side, the smallest triangles are the ones between consecutive lines, and every combination of adjacent smallest triangles (2 together, 3 together, and so on up to all of them) forms another valid, larger triangle sharing that same apex.

- Triangle ABC has two lines drawn from vertex A to points D and E on side BC, with points in order B, D, E, C. Smallest triangles: ABD, ADE, AEC (3 triangles). Pairs of adjacent smallest triangles: ABD+ADE = ABE, and ADE+AEC = ADC (2 more triangles). All three combined: ABC itself (1 more). Total = $3 + 2 + 1 = 6$ triangles.

3. Counting Triangles in a Six-Pointed Star

A six-pointed star (formed by two overlapping equilateral triangles, one pointing up and one pointing down) contains two categories of triangles: the 6 small point triangles at each of the star's outer tips, and the 2 large triangles whose overlap actually forms the star shape — giving a standard total of 8 triangles for the basic figure.

- The 6-pointed star has 6 small tip triangles (one per point) plus the 2 large triangles (the original upward-pointing and downward-pointing triangles whose overlap created the star). Total = $6 + 2 = 8$ triangles. A common mistake is counting only the 6 small tips and forgetting the 2 large overlapping ones that are also complete, valid triangles.

4. Counting All Squares in an $n \times n$ Grid

A grid of $n \times n$ unit squares contains squares of every size from 1×1 up to $n \times n$, and the count of squares of each size $k \times k$ is $(n - k + 1)^2$, so the total number of squares of all sizes is the sum $n^2 + (n - 1)^2 + \dots + 1^2$.

- A 3×3 grid (9 unit squares) contains: 3×3 size squares: $(3 - 3 + 1)^2 = 1$. 2×2 size squares: $(3 - 2 + 1)^2 = 4$. 1×1 size squares: $(3 - 1 + 1)^2 = 9$. Total = $1 + 4 + 9 = 14$ squares of all sizes, not just the 9 visible unit squares.

5. Counting Tilted (Diamond) Squares

When a figure includes lines connecting the midpoints of a square's sides (forming an inner diamond shape), that diamond is itself a valid square (rotated 45°) and must be counted separately from the axis-aligned squares — its sides are equal in length and its angles are all 90° , satisfying the definition of a square despite its tilted orientation.

- A single square has its 4 side-midpoints connected to form an inner diamond. Squares present: the original outer square (1) plus the tilted inner diamond-square (1) = 2 squares total. Overlooking the diamond because it 'looks like a diamond, not a square' is a common undercount.

6. Squares vs. Rectangles: Don't Overcount

When a question specifically asks for the number of SQUARES (not rectangles), every candidate figure must have all 4 sides equal — a grid or divided figure typically also contains many non-square rectangles (like a 1×2 block), and these must be explicitly excluded even though they were 'formed' by the same combination method used to find squares.

- A 1×2 grid of unit squares (2 small unit squares side by side) contains 2 unit squares (1×1 , valid squares) but the overall 1×2 combined region is a rectangle, not a square (unequal sides), and must NOT be added to the square count even though it's a valid combined figure.

7. Counting Both Triangles and Squares in One Composite Figure

Composite figures (a shape built from multiple basic elements together) require running the triangle-count method and the square-count method as two fully separate passes over the same figure, since a line segment can simultaneously be part of both a triangle and a square/rectangle depending on which combination is being counted.

- A rectangle has one diagonal drawn (splitting it into 2 triangles) AND a vertical line down its middle (splitting it into 2 smaller rectangles, each also crossed partially by the diagonal). Triangles: the 2 halves from the diagonal, plus 2 smaller triangles formed where the vertical line crosses the diagonal = 4 triangles total. Rectangles: the 2 halves from the vertical line, plus the original whole rectangle = 3 rectangles total.

8. Worked Full Example: A Pentagon With All Diagonals Drawn

For any unfamiliar composite figure, apply the same universal method regardless of overall shape: label every distinct intersection point, list every smallest enclosed region, then methodically test every combination of 2 or more adjacent smallest regions against the target shape's definition before counting it.

- A pentagon has all 5 of its diagonals drawn, creating a smaller pentagon in the center along with 5 small triangles at the points and 5 slightly larger triangles between them. Counting only 'triangles that touch an outer vertex' undercounts — the full systematic method also finds triangles formed by combining a small point-triangle with an adjacent larger one, giving additional valid triangles beyond the visually obvious ones.

Common Pitfalls

- Counting only the smallest visible units and forgetting the larger figures formed by combining adjacent smaller ones.
- In multi-cevian triangle problems, missing the combination triangles (pairs, triples) and counting only the immediately adjacent smallest triangles.
- In a six-pointed star, counting only the 6 small tip triangles and forgetting the 2 large overlapping triangles that form the star itself.
- Using the $n \times n$ square-counting formula on a grid that isn't actually square (unequal rows and columns), where the formula doesn't directly apply.
- Overlooking tilted (rotated 45° , diamond-oriented) squares formed by midpoint-connecting lines.
- Including non-square rectangles when a question specifically asks only for squares.
- In composite figures, using triangle-counting logic and square-counting logic in a single combined pass instead of two separate, careful passes.

Key Takeaways

- Always count smallest units first, then systematically count every valid combination of adjacent units — never stop at the smallest units alone.
- For triangles sharing one apex, combinations of adjacent smallest triangles (pairs, triples, all together) are themselves valid larger triangles.
- A standard six-pointed star has 8 triangles: 6 small tips plus 2 large overlapping ones.
- An $n \times n$ grid contains $n^2 + (n-1)^2 + \dots + 1^2$ total squares of all sizes, not just the n^2 unit squares.
- Tilted (diamond-oriented) squares from midpoint-connecting lines are still valid squares and must be counted.
- When a question asks for squares specifically, exclude any combined region that is a non-square rectangle.

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IndiaBix — Non-Verbal Reasoning: Construction of Squares and Triangles

<https://www.indiabix.com/non-verbal-reasoning/construction-of-squares-and-triangles/>

GeeksforGeeks — Non-Verbal Reasoning Questions and Answers

<https://www.geeksforgeeks.org/non-verbal-reasoning-questions-and-answers/>